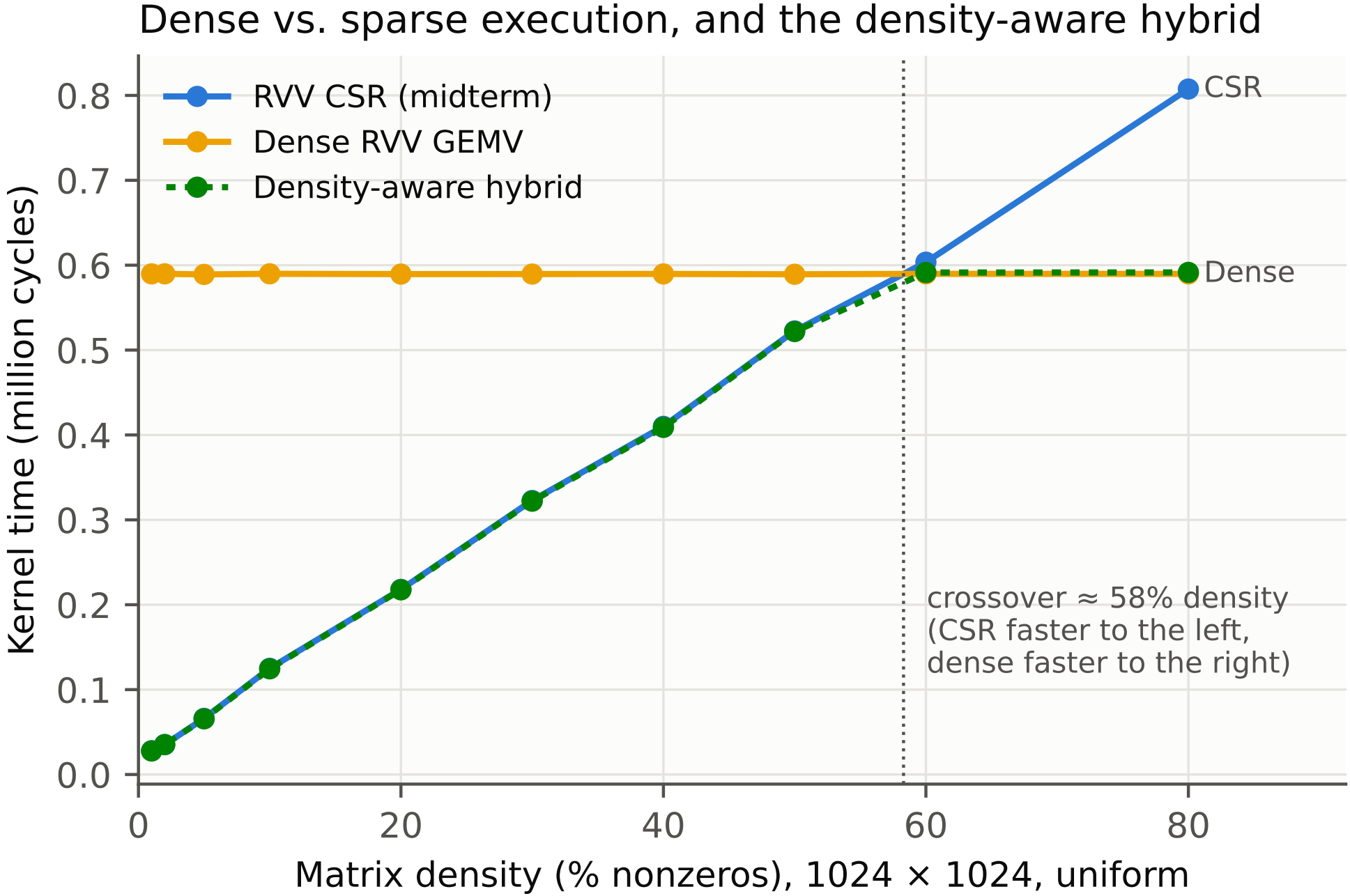
Dense vs. sparse execution, and the density-aware hybrid
RVV CSR (midterm)
Dense RVV GEMV
Density-aware hybrid
CSR
Dense
crossover ≈ 58% density
(CSR faster to the left,
dense faster to the right)
Kernel time (million cycles)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
20
40
60
80
Matrix density (% nonzeros), 1024 × 1024, uniform